HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed Python-based agent orchestration logic on **AWS Lambda**, integrating with **LLM APIs** to process user prompts and generate responses for a customer-facing application.
- Designed and implemented a **FastAPI** service for real-time inference, handling approximately **4,500 requests per hour**, to serve AI-generated content to internal marketing teams.
- Collaborated with product managers and stakeholders to gather requirements for new AI agent features, translating vague business needs into actionable technical specifications for a **lean autonomous team**.
- Built robust data pipelines using **Python** and **SQL** to ingest and preprocess large datasets, preparing them for model training and evaluation on an internal data platform.
- Containerized AI services using **Docker** and managed deployments to a **Kubernetes** cluster, ensuring reliable and scalable operation of production systems.
- Wrote comprehensive unit and integration tests for **Python** and **FastAPI** services, achieving over **85% code coverage** and reducing post-deployment bugs by **30%**.
- Contributed to the design and implementation of **LLM agent orchestration**, focusing on reliability, measurability, and safety for business-critical applications, and participated in technical code reviews.

SKILLS

Languages – Python, SQL, Java, JavaScript, TypeScript, R

Frameworks/Libraries – Spring Boot, FastAPI, Flask, scikit-learn, pandas, NumPy, LangChain, Node.js, NestJS

Cloud/DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git, Linux

Databases – PostgreSQL, MySQL, Oracle

AI/ML – LLM APIs, Agent Orchestration, Prompt Engineering, Model Evaluation, Data Analysis, Machine Learning Fundamentals

BI/Analytics – Power BI, Tableau, DAX, Power Query, Data Modelling

Methodologies – Agile, Scrum, CI/CD

PROJECTS

AI Agent Development for Customer Support Automation

- Engineered a Python-based agent system that integrates with **LLM APIs** to handle customer inquiries, reducing average response time by **40%** for initial triage.
- Developed data ingestion pipelines using **SQL** and **Pandas** to feed customer interaction logs into a feedback loop for agent performance improvement.
- Deployed the agent service as a **Docker** container on an **AWS EC2** instance, establishing basic monitoring with **CloudWatch**.

Marketing Content Generation Pipeline

- Built a **FastAPI** application to serve AI-generated marketing copy based on user-defined prompts, processing approximately **1,000 generation requests daily**.
- Integrated with a **vector database** to store and retrieve relevant context for prompt enhancement, improving content relevance by an estimated **15%**.
- Wrote unit tests for the API endpoints and data processing modules, ensuring functional correctness and reliable output for the marketing team.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)